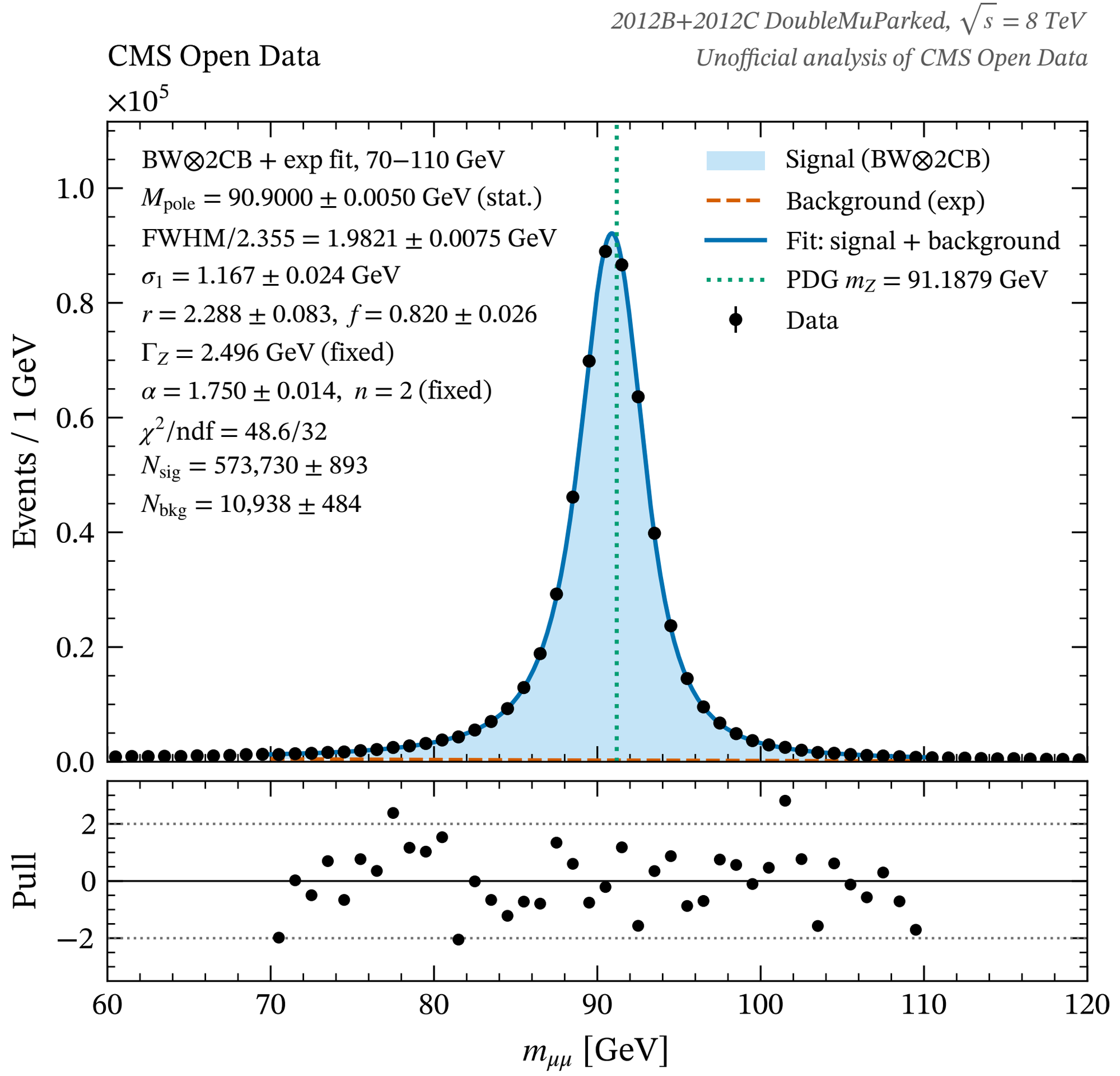

2012B+2012C DoubleMuParked, $\sqrt{s} = 8$ TeV
CMS Open Data
Unofficial analysis of CMS Open Data
$\times 10^5$
BW⊗2CB + exp fit, 70–110 GeV
$M_{\text{pole}} = 90.9000 \pm 0.0050$ GeV (stat.)
FWHM/2.355 = 1.9821 ± 0.0075 GeV
$\sigma_1 = 1.167 \pm 0.024$ GeV
$r = 2.288 \pm 0.083$, $f = 0.820 \pm 0.026$
$\Gamma_Z = 2.496$ GeV (fixed)
$\alpha = 1.750 \pm 0.014$, $n = 2$ (fixed)
$\chi^2/\text{ndf} = 48.6/32$
$N_{\text{sig}} = 573{,}730 \pm 893$
$N_{\text{bkg}} = 10{,}938 \pm 484$
Signal (BW⊗2CB)
Background (exp)
Fit: signal + background
PDG $m_Z = 91.1879$ GeV
Data
Events / 1 GeV
Pull
$m_{\mu\mu}$ [GeV]